Reliable®

Model DDX

Deluge and Preaction

Fire Protection Systems

Operation and Maintenance Manual



Bulletin 520 October 2026
PN 9999970815

www.reliablesprinkler.com/products/system-valves/deluge

THIS PAGE LEFT INTENTIONALLY BLANK

Contents

IMPORTANT: This manual is provided to assist with the installation, operation, and maintenance of Reliable Automatic Sprinkler Company Model DDX deluge and preaction systems. It is imperative that this manual be completely read and understood before installation or use of the product. Save this manual for future reference.

1	General	Page 1
2	Model DDX System Types and References	Page 1
3	Setup Procedure	Pages 2-4
4	Commissioning	Page 5
5	Inspection	Page 5
6	Testing	Pages 5-7
7	Maintenance	Page 8
8	Troubleshooting	Pages 9-10
9	Repair Procedures	Pages 11-12
10	Supplemental Information	Pages 13-14
11	Additional Technical Support & Notes	Page 15
12	Annex A1: Commissioning Checklist	Pages 16-17
13	Annex A2: Model DDX Deluge Valve Components	Page 18
14	Annex A3: Model DDX Deluge Parts List	Pages 19-21

1. General

This manual is provided to assist with the operation, inspection, testing, and maintenance of the Reliable Automatic Sprinkler Company Model DDX deluge valve and related fire protection systems. It is imperative that this manual be completely read and understood before installation or use of the product. Save this manual for future reference. Please note the following important information:

- Failure to comply with this information in this document can cause failure of the system resulting in serious personal injury and property damage.
- The fire protection system will be out of service until the setup procedure is completed.
- Notify any necessary authorities having jurisdiction and other affected personnel prior to placing the fire protection system out of service.
- This procedure should only be performed by persons who are experienced and trained in the installation and operation of water-based fire protection systems.
- This document does NOT supersede the requirements of NFPA standards nor those of the authorities having jurisdiction.

2. Model DDX System Types and References

Please reference the attached resources prior to commencing work on the system.

- **Deluge System:** Intended for use where the rapid application of water or fire-fighting foam from open discharge devices is necessary. Operation of the deluge valve occurs upon activation of an automatic detection system or opening of the emergency manual release. Please reference DDX Deluge Valve | Reliable Sprinkler and Technical Bulletin 519.

- **Low-pressure Dry System:** Intended for use where a wet system cannot be utilized due to possible or known freezing conditions. Operation of the deluge valve occurs upon loss of system pneumatic pressure or opening of the emergency manual release. Please reference DDX-LP Low Pressure Dry Pipe Valve | Reliable Sprinkler and Technical Bulletin 338.

- **Single Interlock Preaction System:** Intended for use where the accidental release of water from automatic fire sprinklers in the protected area must be avoided. Operation of the deluge valve occurs upon activation of an automatic detection system or opening of the emergency manual release. Please reference DDX Single Interlock Preaction Systems | Reliable Sprinkler and Technical Bulletin 749.

- **Double Interlock Preaction System:** Intended for use where the accidental release of water from automatic fire sprinklers in the protected area must be avoided. Operation of the deluge valve occurs upon activation of an automatic detection system and loss of system pneumatic pressure, or opening of the emergency manual release. Please reference DDX Double Interlock Preaction System Type D - Electric/Electric | Reliable Sprinkler and Technical Bulletins 750 (Type D Electric/Electric), DDX Double Interlock Preaction System Type F - Electric/Pneumatic | Reliable Sprinkler and Technical Bulletin 751 (Type F Electric/Pneumatic), or DDX Double Interlock Preaction System Type PL - Pneumatic/Pneumatic | Reliable Sprinkler and Technical Bulletin 752 (Type PL Pneumatic/Pneumatic).

- **Non-interlock System:** Intended for use where electric release single interlock preaction is the primary manner of protection and where the system will operate in fail-safe mode as a low-pressure dry system (see above). Please reference Model DDX Non Interlock Preaction System | Reliable Sprinkler and Technical Bulletin 756.

3. Setup Procedure

Note: Some steps may not be necessary for initial setup, however, confirm that trim valves are in proper open or closed position.

1. Close the valve controlling water supply to the system.
2. Close the valve supplying hydraulic pressure to the pushrod chamber.
3. Close the valve supplying pneumatic pressure to the system and dry pilot line (if present).
4. Open the emergency manual release valve.
5. Open main drain valve and drain system.
6. Drain all low points throughout the system. Make sure that all drain valves are closed and/or plugs are replaced.
7. If necessary, repair or replace any damaged fire sprinkler system components.
8. Verify that the alarm line ball valve is open and push in the plunger of ball drip to drain the intermediate chamber of the Model DDX valve.
9. Push in the external reset knob on the back of the Model DDX valve. While holding the reset knob in, turn the knob counterclockwise (as you face the valve). This will rotate the lever and allow the clapper to reset.

Important! The pushrod chamber must be fully vented and drained when resetting the clapper.

10. Verify that the automatic release mechanism (wet pilot line, dry pilot actuator, or solenoid valve) is open.
11. Open the valve supplying hydraulic pressure to the pushrod chamber and allow water to fill the pushrod chamber.
12. Slowly close the emergency manual release valve to direct water to the automatic release mechanism.

Proceed below based upon the type of system, then proceed to step 13:

Deluge System, Wet Pilot Line Actuation Trim (also reference Technical Bulletin 519)

12a. Bleed the entire wet pilot line until all air is removed. Close the opening on the wet pilot line when there is a solid stream of water.
12b. Confirm pressurization of the wet pilot line and pushrod chamber on the pushrod chamber water pressure gauge.

Deluge System, Dry Pilot Line Actuation Trim (also reference Technical Bulletin 519)

12a. Bleed all air from the water side of the pneumatic actuator. When there is a solid stream of water from the drain outlet, apply compressed air or nitrogen pressure to the pilot line actuator causing it to close.
12b. Confirm pressurization of the pushrod chamber on the pushrod chamber water pressure gauge.
12c. Confirm pressurization of the dry pilot line on the dry pilot line air pressure gauge.
12d. Adjust the dry pilot line air or nitrogen supply pressure to the appropriate value as shown in Table A on page 13.

Deluge System, Electric Actuation Trim (also reference Technical Bulletin 519)

12a. When there is a solid stream of water from the solenoid valve, reset the detection system causing the solenoid valve to close.
12b. Confirm pressurization of the pushrod chamber on the pushrod chamber water pressure gauge.

Low Pressure (LP) Dry System (also reference Technical Bulletin 338)

12a. Bleed all air from the water side of the pneumatic actuator. When there is a solid stream of water from the drain outlet, apply compressed air or nitrogen pressure to the pilot line actuator causing it to close.
12b. Confirm pressurization of the pushrod chamber on the pushrod chamber water pressure gauge.
12c. Confirm pressurization of the dry pilot line on the dry pilot line air pressure gauge.
12c. Open the valve supplying pneumatic pressure to the system. Adjust the system air or nitrogen pressure to the appropriate value as shown in Table A on page 13 and place the pneumatic supply into automatic operation.

3. Setup Procedure (cont.)

Single Interlock Preaction System, Wet Pilot Line Actuation Trim (also reference Technical Bulletin 749)

12a. Bleed the entire wet pilot line until all air is removed. Close the opening on the wet pilot line when there is a solid stream of water.
12b. Confirm pressurization of the wet pilot line and pushrod chamber on the pushrod chamber water pressure gauge.
12c. Open the valve supplying pneumatic pressure to the system. Adjust the system air or nitrogen pressure to 15 psi (1.03 bar) and place the pneumatic supply into automatic operation.

Single Interlock Preaction System, Dry Pilot Line Actuation Trim (also reference Technical Bulletin 749)

12a. Bleed all air from the water side of the pneumatic actuator. When there is a solid stream of water from the drain outlet, apply compressed air or nitrogen pressure to the pilot line actuator causing it to close.
12b. Confirm pressurization of the pushrod chamber on the pushrod chamber water pressure gauge.
12c. Confirm pressurization of the dry pilot line on the dry pilot line air pressure gauge.
12d. Adjust the dry pilot line air or nitrogen supply pressure to the appropriate value as shown in Table F on page 4.
12e. Open the valve supplying pneumatic pressure to the system. Adjust the system air or nitrogen pressure to 15 psi (1.03 bar) and place the pneumatic supply into automatic operation.

Single Interlock Preaction System, Electric Actuation Trim (also reference Technical Bulletin 749)

12a. When there is a solid stream of water from the solenoid valve, reset the detection system causing the solenoid valve to close.
12b. Confirm pressurization of the pushrod chamber on the pushrod chamber water pressure gauge.
12c. Open the valve supplying pneumatic pressure to the system. Adjust the system air or nitrogen pressure to 15 psi (1.03 bar) and place the pneumatic supply into automatic operation.

Double Interlock Preaction System, Type D Electric/Electric (also reference Technical Bulletin 750)

12a. When there is a solid stream of water from the solenoid valve, reset the detection system causing the solenoid valve to close.
12b. Confirm pressurization of the pushrod chamber on the pushrod chamber water pressure gauge.
12c. Open the valve supplying pneumatic pressure to the system. Adjust the system air or nitrogen pressure to 15 psi (1.03 bar) and place the pneumatic supply into automatic operation.

Double Interlock Preaction System, Type F Electric/Pneumatic (also reference Technical Bulletin 751)

12a. Bleed all air from the water side of the pneumatic actuator. When there is a solid stream of water from the drain outlet, apply compressed air or nitrogen pressure to the pilot line actuator causing it to close.
12b. Reset the detection system causing the solenoid valve to close.

Important! Do NOT reverse steps 12a and 12b. Water must be moving through the solenoid valve and the LP actuator prior to step 12a. Always close the pneumatic actuator before closing the solenoid valve.

12c. Confirm pressurization of the pushrod chamber on the pushrod chamber water pressure gauge.
12d. Open the valve supplying pneumatic pressure to the system. Adjust the system air or nitrogen pressure to the appropriate value as shown in Table A on page 13 and place the pneumatic supply into automatic operation.

3. Setup Procedure (cont.)

Double Interlock Preaction System, Type PL Pneumatic/Pneumatic (also reference Technical Bulletin 752)

12a. Bleed all air from the water side of the pneumatic actuator. When there is a solid stream of water from the drain outlet, apply compressed air or nitrogen pressure to the pilot line causing the LP actuator to close.
12b. Confirm pressurization of the pushrod chamber on the pushrod chamber water pressure gauge.
12c. Adjust the dry pilot line air or nitrogen pressure to the appropriate value as shown in table X on page X and place the pneumatic supply into automatic operation.
12d. Open the valve supplying pneumatic pressure to the system. Adjust the system air or nitrogen pressure to the appropriate value as shown in Table A on page 13 and place the pneumatic supply into automatic operation.

Non-Interlock System (also reference Technical Bulletin 756)

12a Bleed all air from the water side of the pneumatic actuator. When there is a solid stream of water from the drain outlet, apply compressed air or nitrogen pressure to the pilot line actuator causing it to close.
12b. Bleed all air from the water side of the solenoid valve. When there is a solid stream of water from the solenoid valve, reset the detection system causing it to close.
12c. Confirm pressurization of the pushrod chamber on the pushrod chamber water pressure gauge.
12d. Open the valve supplying pneumatic pressure to the system. Adjust the system air or nitrogen pressure to the appropriate value as shown in Table A on page 13 and place the pneumatic supply into automatic operation.

Continue for all system types:

13. Slowly open the valve controlling water supply to the system while simultaneously closing the main drain valve.
14. Once the main drain valve is fully closed, ensure that the main control valve is fully open. Verify system water supply pressure on the system water pressure gauge.
15. Secure the emergency manual release valve with a cable tie to prevent accidental release.
16. Reset all alarms and notify all necessary authorities having jurisdiction and other affected personnel that the system has been returned to service.

4. Commissioning

The document in Annex A is provided to assist in the commissioning of Model DDX systems. It is not intended to replace any testing and/or documentation required by NFPA standards (e.g., Contractor's Material and Test Certificate), contract documents, or the authorities having jurisdiction.

5. Inspection

1. Confirm that the valve controlling water supply to the system is OPEN and properly monitored.
2. Confirm the manual emergency release valve is CLOSED and secured in that position.
3. Confirm that the pushrod chamber supply valve is OPEN and the pushrod chamber is hydraulically pressurized. Note that pushrod chamber pressure should equal or exceed the system water pressure.
4. Confirm the alarm line valve is OPEN.
5. Confirm the condensate drain valve is CLOSED.
6. Confirm the alarm test valve is CLOSED.
7. Confirm there is no leakage present from the weep hole on the bottom of the pushrod chamber.
8. Confirm that the air or nitrogen pressure supply valves are OPEN and 15 psi (1.03 bar) supervisory pressure is being maintained on the system (excluding deluge systems).
9. If present, confirm that dry pilot line air or nitrogen pressure supply valves are OPEN and supervisory pressure conforms to Table X of this document.
10. Confirm there is no leakage from the releasing trim and inspect all releasing devices for leakage. Verify that tubing drain lines from releasing devices are not pinched or crushed which could prevent proper activation of the deluge valve.
11. Push the plunger on the velocity check valve to be sure ball is off its seat. If no water appears, the deluge valve water seat is tight.

6. Testing

Alarm Testing

1. Water flow: Open the alarm test valve permitting water from the supply to flow to the electric sprinkler alarm switch and/or the mechanical sprinkler alarm (water motor). Confirm proper alarms are reported. After testing, close alarm test valve securely. Push in on the plunger of ball drip valve and hold until all water has drained from the alarm line.
2. Low pneumatic pressure: Close the valve controlling air or nitrogen supply to the system or dry pilot line. Slowly open a valve or port on the section being tested. Note the pressure at which the pressure switch operates. Close the valve or port used for testing and return the pneumatic pressure system to automatic operation.
3. High pneumatic pressure (if so equipped): Slowly increase the pneumatic pressure being supplied to the section being tested. Note the pressure at which the pressure switch operates. Slowly open a valve or port on the section being tested and lower the pneumatic pressure to the proper pressure. Close valve or port when proper pressure has been achieved and return the pneumatic pressure system to automatic operation.

6. Testing (cont.)

Partial Operational Test

1. Close the valve controlling water supply to the deluge valve and open the main drain valve.
2. Close the valve supplying hydraulic pressure to the pushrod chamber. Confirm that hydraulic pressure on the pushrod chamber remains stable. Loss of pressure in the pushrod chamber indicates failure of the check valves in the pushrod chamber supply line.
3. Open the valve supplying hydraulic pressure to the pushrod chamber.
4. Open the emergency manual release valve. Confirm a sudden drop in pushrod chamber hydraulic pressure occurs.
5. Close the emergency manual release valve and confirm hydraulic pressure in the pushrod chamber returns to match incoming water supply pressure.
6. Slowly open the valve controlling water supply to the system while simultaneously closing the main drain valve.
7. Once the main drain valve is fully closed, ensure that the main control valve is fully open. Verify system water supply pressure on the system water pressure gauge.
8. Secure the emergency manual release valve with a cable tie to prevent accidental release.
9. Reset all alarms and notify all necessary authorities having jurisdiction and other affected personnel that the system has been returned to service.

Full Operational Test: Model DDX LP Deluge System
Caution! THIS TEST WILL CAUSE THE DELUGE VALVE TO OPEN AND ACTIVATE WATER FLOW ALARMS.

1. Operate the automatic detection system (wet pilot line, dry pilot line, or electrical device).
2. Confirm a sudden drop in the pushrod chamber hydraulic pressure and activation of the deluge valve.
3. Confirm proper water flow alarm activation.
4. Close main control valve to stop water flow to the system.
5. Reset the system following the instructions in this document. Secure the manual emergency release valve in the OFF position with nylon tie after deluge valve is reset.

Full Operational Test: Model DDX LP Dry System
Caution! THIS TEST WILL CAUSE THE DELUGE VALVE TO OPEN AND ACTIVATE WATER FLOW ALARMS.

1. Disable the automatic air or nitrogen supply to the system.
2. Open a test valve or port on the system to release pneumatic pressure.
3. Confirm a sudden drop in the pushrod chamber hydraulic pressure and activation of the deluge valve.
4. Confirm proper water flow alarm activation.
5. Close main control valve to stop water flow to the system.
6. Reset the system following the instructions in this document. Secure the manual emergency release valve in the OFF position with nylon tie after deluge valve is reset.

Full Operational Test: Model DDX Single Interlock Preaction System
Caution! THIS TEST WILL CAUSE THE DELUGE VALVE TO OPEN AND ACTIVATE WATER FLOW ALARMS.

1. Operate the automatic detection system (wet pilot line, dry pilot line, or electrical device).
2. Confirm a sudden drop in the pushrod chamber hydraulic pressure and activation of the deluge valve.
3. Confirm proper water flow alarm activation.
4. Close main control valve to stop water flow to the system.
5. Reset the system following the instructions in this document. Secure the manual emergency release valve in the OFF position with nylon tie after deluge valve is reset.

6. Testing (cont.)

Full Operational Test: Model DDX Type D and Type F Double Interlock Preaction System
Caution! THIS TEST WILL CAUSE THE DELUGE VALVE TO OPEN AND ACTIVATE WATER FLOW ALARMS.

1. Operate the automatic detection system.
2. Confirm receipt of detection system alarm. DELUGE VALVE SHOULD NOT ACTIVATE.
3. Disable the automatic air or nitrogen supply to the system.
4. Open a test valve or port on the system to release pneumatic pressure.
5. Confirm a sudden drop in the pushrod chamber hydraulic pressure and activation of the deluge valve.
6. Confirm proper water flow alarm activation.
7. Close main control valve to stop water flow to the system.
8. Reset the system following the instructions in this document. Secure the manual emergency release valve in the OFF position with nylon tie after deluge valve is reset.

Full Operational Test: Model DDX Type PL Double Interlock Preaction System
Caution! THIS TEST WILL CAUSE THE DELUGE VALVE TO OPEN AND ACTIVATE WATER FLOW ALARMS.

1. Disable the automatic air or nitrogen supply to the dry pilot line.
2. Open a test valve or port on the dry pilot line to release pneumatic pressure.
3. Confirm receipt of detection system alarm. DELUGE VALVE SHOULD NOT ACTIVATE.
4. Disable the automatic air or nitrogen supply to the system.
5. Open a test valve or port on the system to release pneumatic pressure.
6. Confirm proper water flow alarm activation.
7. Confirm a sudden drop in the pushrod chamber hydraulic pressure and activation of the deluge valve.
8. Close main control valve to stop water flow to the system.
9. Reset the system following the instructions in this document. Secure the manual emergency release valve in the OFF position with nylon tie after deluge valve is reset.

Full Operational Test: Model DDX Non-Interlock System
Caution! THIS TEST WILL CAUSE THE DELUGE VALVE TO OPEN AND ACTIVATE WATER FLOW ALARMS.

1. Follow steps for Full Operational Test: Model DDX LP Dry System.
2. Reset the system following the instructions in this document.
3. Follow steps for Full Operational Test: Model DDX Single Interlock Preaction System.

7. Maintenance

General

1. The owner is responsible for maintaining the fire protection system in proper operating condition. Any system maintenance or testing that involves placing a control valve or detection/control system out of service may eliminate the fire protection that is provided by the fire protection system.
2. The Reliable Model DDX valve and associated equipment shall periodically be given a thorough inspection and test. NFPA 25, “Inspection, Testing, and Maintenance of Water Based Fire Protection Systems,” provides minimum requirements. The information provided in this document does not supersede the minimum requirements of NFPA 25 and/or other owner requirements.
3. System components shall be tested, operated, cleaned, and inspected at least annually, and parts replaced as required. Replace any components found to be corroded, damaged, worn, or non-operable. Increase the frequency of inspections when the valve is exposed to corrosive conditions or chemicals that could impact materials or operation of the assembly.
4. If face plate is removed during maintenance, torque face plate bolts to the following values during re-installation:
   35 ft-lbs. (47 N-m) for 2” through 4” valves
   70 ft-lbs. (95 N-m) for 6”-8” valves

Draining Excess/Condensate Water from above the Model DDX Deluge Valve

1. Notify the owner and monitoring company that maintenance is being performed on the system.
2. Close the main water control valve.
3. Open the Main Drain Valve.
4. Open the Condensate Drain Valve until all water has drained.
5. Close Condensate Drain Valve.
6. Partially open the Main Water Control Valve.
7. Slowly close the Main Drain Valve.
8. Fully open the Main Water Control Valve.
9. Notify the owner and monitoring company that system has been returned to service.

8. Troubleshooting

Symptom	Possible Cause	Action
Water flow alarms not operating.	Malfunctioning alarm device.	Inspect and repair water motor alarm and/or pressure switch.
	Improper pressure switch setting.	Adjust pressure switch.
	Incorrect pressure switch.	Replace with correct switch (PS-10)
	Insufficient water pressure/flow in alarm line.	Inspect and clear waterway between clapper seat and alarm device(s).
False water flow alarm.	Alarm test valve is open or leaking by.	Close or repair alarm test valve.
	Water leakage from below Model DDX clapper into intermediate chamber.	Inspect and repair clapper seal assembly. Inspect and repair lower seat O-ring seal (not common).
	Alarm line valve left closed.	Open alarm line valve.
	Velocity (ball) check closed.	Push plunger to release water.
Low or high pneumatic pressure alarms not operating.	Malfunctioning pressure switch.	Inspect and repair pressure switch.
	Improper pressure switch setting.	Adjust pressure switch.
	Incorrect pressure switch.	Replace with correct switch (PS-15 or PS-25)
Pushrod chamber hydraulic pressure not being maintained.	Pushrod chamber supply valve closed.	Open and leave open pushrod chamber supply valve.
	Leaks in release trim.	Inspect and repair trim.
	Pushrod chamber supply line check valve failure.	Inspect and replace as necessary.
	Clogged strainer on supply line to pushrod chamber.	Clear strainer screen.
	Clogged restricted orifice on pushrod cover inlet.	Disassemble, inspect and clean.
	Faulty gauge reading.	Replace gauge.
Pushrod chamber hydraulic pressure not dropping during operational test.	Plugged or restricted release devices.	Inspect and clear devices.
	Kinked or plugged discharge hoses.	Inspect, clear, and realign as needed.
	Malfunctioning solenoid.	Inspect for proper operation, voltage, etc. Replace as needed.
Excessive pushrod chamber hydraulic pressure.	Thermal expansion.	Install pushrod chamber pressure relief kit.
	Supply water pressure surges.	Install pushrod chamber pressure relief kit.
	Faulty gauge reading.	Replace gauge.
Water leakage from pushrod chamber weep hole.	Compromised pushrod diaphragm.	Inspect and replace.
Pneumatic pressure leakage from pushrod chamber weep hole.	Compromised O-ring inside pushrod guide.	Inspect and replace.
	Compromised O-ring on pushrod guide threads.	Inspect and replace.

8. Troubleshooting (cont.)

Symptom	Possible Cause	Action
Sluggish deluge valve operation.	Insufficient hydraulic pressure release.	See “Pushrod chamber hydraulic pressure not dropping during operational test.”
	Pushrod not moving freely.	Inspect, clean, and lubricate.
Clapper not resetting properly.	Pressure on pushrod chamber.	Open emergency manual release valve to release pressure.
	Debris in valve body.	Remove cover plate, inspect and clean.
Deluge valve not activating properly during operational test.	Insufficient hydraulic pressure release.	See “Pushrod chamber hydraulic pressure not dropping during operational test.”
	Excessive pneumatic pressure above clapper.	Adjust pneumatic pressure to conform to required pressure. Adjust low and high pressure switch settings accordingly.
	Debris inside valve body.	Remove cover plate, inspect and clean.
Continuous water loss through alarm line velocity check.	Water leakage from below Model DDX clapper into intermediate chamber.	Inspect and repair clapper seal assembly. Inspect and repair lower seat O-ring seal (not common).
	Alarm test valve not fully closed, or leaking past	Fully close or replace alarm test valve.
Intermittent water loss through alarm line velocity check.	Residual water on top of clapper leaking into intermediate chamber.	Inspect clapper seal and replace if necessary. Inspect and repair upper seat O-ring seal (not common).
Continuous air or nitrogen loss through alarm line out of velocity check.	System pneumatic pressure leaking into intermediate chamber.	Inspect clapper seal and replace if necessary. Inspect and repair upper seat O-ring seal (not common).

9. Repair Procedures

Model DDX Deluge Valve Clapper

1. Notify any necessary authorities having jurisdiction and other affected personnel prior to the performance of any work.
2. Disable detection system, if present.
3. Close the valve controlling the water supply to the deluge valve and open the main drain valve. Open the condensate drain valve. Close the pushrod chamber supply valve and open the emergency manual release valve.
4. Remove the deluge valve face plate. Inspect the seat, clapper, and seal assembly for damage.
5. If damage to the clapper seal assembly is found, replace as follows:
   a. Unthread the bumpstop nuts and remove the seal assembly. Install a new seal assembly and thread the bumpstop nuts onto the threaded studs of the seal assembly. Tighten finger tight plus ¼ to ½ turn.
6. If damage to the clapper is found, replace as follows:
   a. Disconnect the condensate drain trim section from the back of the valve.
   b. Remove the retaining rings from the clapper hinge pin and push the hinge out through the condensate drain opening. Remove the clapper subassembly.
   c. Install a new clapper subassembly in the reverse order making sure the clapper spacers are in their proper position.
   d. Reinstall condensate drain.
7. If damage to the seat is found, or it is suspected that there is leakage past the seat O-rings, replace as follows:
   a. Using Reliable P/N 6881603000 Seat Wrench for 2" (50mm), 2½" (65mm), 76mm and 3" (80mm) valve sizes, Reliable P/N 6881604000 for 4" (100mm) valve size, Reliable P/N 6881606000 for the 6" (150mm) and 165mm valve sizes or Reliable P/N 6881608000 Seat Wrench for 8" (200mm) valve size, unthread the seat from the valve body and remove it from the valve.
   b. Remove the clapper-mounting ring subassembly from the valve.
   c. Visually examine all components and replace any component that appears damaged.
   d. Prior to reassembly, clean the bore of the valve body and lubricate the bore with O-ring grease.
   e. Lubricate and install new O-rings onto the seat. Lubricate and install the mounting ring O-ring into the body (8" (200mm) valve size only).
   f. Position the clapper-mounting-ring subassembly over the opening in the valve body. Use caution to not damage or dislodge the mounting ring O-ring (8" (200mm) valve size only).
   g. Align the mounting ring so that the lever/latch assembly faces the pushrod and the mounting ring ears are between the tabs on the floor of the valve body. Insert the seat into the valve body and through the clapper-mounting ring subassembly.
   h. Thread the seat into the body by hand, then tighten the seat with the seat wrench until secure. Verify that the seat-clapper-mounting ring subassembly is flush with the floor of the valve body, the mounting ring ears are centered between the tabs of the body, and the lever/latch lines up with the pushrod.
8. Inspect the face plate gasket for any damage and replace if necessary.
9. Reinstall the deluge valve face plate and set up the Model DDX Deluge Valve following the Setup Procedure in this document.

Model DDX Deluge Valve Pushrod Chamber

1. Locate the small weep hole on the underside of the pushrod chamber and inspect for water or air/nitrogen leakage.
2. Water leakage from the weep hole is caused by a ruptured pushrod diaphragm. Replace as follows:
   a. Notify any necessary authorities having jurisdiction and other affected personnel prior to the performance of any work.
   b. Disable detection system, if present.
   c. Close the valve controlling the water supply to the deluge valve and open the main drain valve. Open the condensate drain valve. Close the pushrod chamber supply valve and open the emergency manual release valve.
   d. Remove the trim at the unions nearest to the pushrod chamber cover.
   e. Remove the pushrod chamber cover by unthreading the six retaining screws.
   f. Visually inspect the pushrod chamber cover and piston to determine what might have damaged the diaphragm. Correct any issues found.
   g. Place a new diaphragm over the push rod.

9. Repair Procedures (cont.)

IMPORTANT: The diaphragm has two different surfaces. It is not bi-directional and will fail if installed incorrectly.





If necessary, invert the diaphragm so that the fabric side covers the pushrod and the smooth (lettered) surface conforms to the inside of the pushrod chamber cover. When properly oriented, the fabric side is to the right when facing the valve, and the smooth side is to the left.

h. Reinstall the cover using the six retaining screws. Tighten to a torque of 15 foot-pounds in a star pattern.
i. Set up the Model DDX Deluge Valve following the Setup Procedure in this document.

3. Air or nitrogen leakage from the weep hole is caused by compromised O-rings where the pushrod penetrates the valve body. Replace as follows:
   a. Disassemble following steps a through e above.
   b. Gently pull the pushrod out of the nylon bushing in the valve body.
   c. Unthread nylon bushing and remove.
   d. Remove external and internal O-rings from the bushing. Lubricate new O-rings and install into nylon bushing.
   e. Reinstall nylon bushing into valve body to a hand-tight state.
   f. If necessary, clean and lubricate pushrod. Reinstall through nylon bushing.
   g. Reassemble following steps g through l above. Use of a new pushrod diaphragm is recommended.

10. Supplemental Information

Pneumatic Pressure Requirements

Unless revised herein, all requirements of the current edition of NFPA 13 shall apply to the components and systems that supply pneumatic pressure to Reliable Model DDX systems requiring pneumatic supervisory or operational pressure.

Please note that tank-less compressors where pressure is controlled by the on-off settings of a pressure switch are strongly discouraged where accelerators are present.

Reliable Model DDX single-interlock, Type D double-interlock, and Non-interlock preaction systems are designed to operate with a system supervisory pressure of 15 psi (1.0 bar).

Reliable Model DDX systems using a Model LP or PL pneumatic actuator, including dry pilot line detection systems, must have a prescribed amount of air or nitrogen pressure to offset the incoming water supply. The minimum pneumatic pressure required is provided in Table A below.

System Pneumatic Pressure

Maximum Water Pressure psi (bar)	Minimum Air/Nitrogen Pressure psi (bar)
20 (1.4)	8 (0.6)[2]
30 (2.1)	10 (0.7)[2]
50 (3.4)	12 (0.8)[2]
75 (5.2)	13 (0.9)
100 (6.9)	15 (1.0)
125 (8.6)	16 (1.1)
150 (10.3)	17 (1.2)
175 (12.1)	18 (1.2)
200 (13.8)	19 (1.3)
225 (15.5)	21 (1.4)
250 (17.2)	22 (1.5)
275 (19.0)	23 (1.6)
300 (20.7)	24 (1.7)

Notes:
1. Pneumatic pressure less than the minimum specified will result in inadvertent operation of the system.
2. Pneumatic pressure higher than the minimum specified may result in delayed operation of the system.
Caution! Where the maximum water pressure is 20 psi (1.4 bar), 30 psi (2.1 bar) or 50 psi (3.4 bar) the maximum pneumatic pressure cannot exceed the 10 psi (.7 bar) 14 psi (1 bar) and 16 psi (1.1 bar) respectively. This may prohibit the use of a tank-less air compressor (see note 4).
3. The use of a listed pressure maintenance device such as the Reliable Model A or Model DD in conjunction with a tank-mounted air compressor or nitrogen generator incorporating a tank is highly recommended, even for single systems. (This is an NFPA 13 requirement for multiple systems fed from a common source.)
4. Where a tank-less compressor is used, the “start/on” setting of the pressure switch shall not be less than the minimum pneumatic pressure in the table. The “stop/off” pressure will be determined by the required differential of the compressor pressure switch.
5. During set-up, higher pneumatic pressure may temporarily be required to set the Model LP Actuator. Following set-up, lower the pneumatic pressure of the system to be in accordance with the table.
6. In no case should the pneumatic pressure exceed 30 psi (2.1 bar) to prevent damage to the actuator diaphragm.

LP Dry Pilot Actuator Characteristics

Maximum Water Pressure psi (bar)	Nominal Trip Point psi (bar)
20 (1.4)	5 (0.36)
30 (2.1)	5 (0.36)
50 (3.4)	7 (0.51)
75 (5.2)	9 (0.63)
100 (6.9)	10 (0.72)
125 (8.6)	11 (0.78)
150 (10.3)	12 (0.83)
175 (12.1)	13 (0.9)
200 (13.8)	13 (0.9)
225 (15.5)	14 (1)
250 (17.2)	14 (1)
275 (19.0)	14 (1)
300 (20.7)	14 (1)

Note:
1. The Dry Pilot Actuator differential should be used as the “Dry Valve Differential” for valves using the Reliable Model LP Dry Pilot Actuator. Actual trip points may vary ± 2 psi due to the design of the actuator.

10. Supplemental Information (cont.)

Solenoid Valves

Reliable Model DDX systems that incorporate an electric release component use a ½" NPT normally closed (energize to open) 24 VDC solenoid valve with either a 175 psi (12.1 bar) or 300 psi (20.7 bar) pressure rating.

Standard Solenoid Valve Specifications:
Parker-Hannifin Model 73218BN4UNLVN0C111C2
Rated working pressure: 175 psi (12.1 bar)
Voltage: 24 VDC
Power: 10 Watts
Current: 0.41 Amps Holding
Enclosure Coil: NEMA 4X
Pipe Size: ½" NPT Female
Cv Factor: 4.0

Alternate Solenoid Valve Specifications:
Parker-Hannifin Model 73212BN4TNLVN0C322C2
Rated working pressure: 300 psi (20.7 bar)
Voltage: 24 VDC
Power: 22 Watts
Current: 0.83 Amps Holding
Enclosure Coil: NEMA 4X
Pipe Size: ½" NPT Female
Cv Factor: 2.8

Faulty solenoid valves should be replaced. Disassembly and repair of the solenoid valve is not recommended and should be done only by a trained technician. An improperly repaired or partially assembled solenoid valve could result in accidental operation of the valve or failure of the valve to operate.

Accelerators

The Reliable Model B1 accelerator is approved for use on the following system types:
Model DDX LP Dry System
Model DDX Type F Double Interlock
Model DDX Type PL Double Interlock
Model DDX Non-Interlock

Please refer to Technical Bulletin 323 for additional information on the Model B1 Accelerator. When present, it is important that the accelerator be fully serviced in accordance with the bulletin after each operation. The accelerator shall be isolated from the system during the setup procedure and placed into service only after the pneumatic pressure has stabilized and the pneumatic supply system has been placed into automatic operation. Note that the Model B1 accelerator requires a minimum air or nitrogen pressure of 15 psi (1.03 bar) for proper operation. The use of tankless air compressors is NOT recommended when utilizing the Model B1 accelerator.

Additional Technical Support

Reliable Product Technology department may be reached at 800-557-2726, or via email at techserv@reliablesprinkler.com.

Notes:

Annex A1

Reliable Model DDX System Commissioning Checklist

Note: This document is intended to assist in the commissioning of the Reliable Model DDX systems. It is NOT intended to supersede the requirements of NFPA 13, contract documents, or requirements of the authority having jurisdiction. Compliance with ALL requirements is the responsibility of the installing contractor.

Notify the building owner, occupants, and central station monitoring company that testing is being performed. Fully read each step prior to performing work. Record requested information and initial each item upon completion.

1. Following acceptance of the installation and completion of hydrostatic testing, place the system into operational status.

Record water supply pressure and supervisory air pressure: _____

2. Begin closing the main water supply control valve. Confirm that the valve tamper supervisory signal is activated. Fully open the main water supply control valve to clear the valve supervisory signal.

Valve tamper supervisory signal operates properly: ☐

3. Disable the air or nitrogen supply to the system. Slowly open the condensate drain valve on the back of the Model DDX deluge valve. Confirm that the low air supervisory signal is activated. Record pressure at which the low air signal occurs. Return the air or nitrogen supply to normal operation to clear the low pressure signal.

Low air supervisory signal operates properly at _____ psi:

4. Slowly increase the air or nitrogen pressure on the system. Confirm that the high air supervisory signal is activated. Record pressure at which the high air signal occurs. Open the condensate drain to reduce pressure to appropriate level to clear the high pressure signal.

High air supervisory signal operates properly at _____ psi:

5. Open the water flow alarm test valve. Confirm that water flow alarm signal is activated. Close the water flow alarm test valve and press the plunger on the velocity check valve to clear the water flow alarm signal.

Water flow alarm test operates properly: ☐

6. Activate detection system. Confirm that detection and alarm signals are activated. For single interlock preaction systems, activation of the detection system will result in immediate system release and subsequent water flow alarm signal.

Detection and alarm signals operate properly. For single interlock preaction system, deluge valve operates, and water flow alarm signal operates properly: ☐

For double interlock preaction systems, leave detection system in active state and proceed to step 7.

Annex A1

Reliable Model DDX System Commissioning Checklist

7. Close the air supply valve on the pressure maintenance device. Slowly open the condensate drain valve on the lower back side of the Model DDX deluge valve. Loss of air pressure during an active detection signal will result in system release and subsequent water flow alarm signal.

Double interlock preaction system operates properly upon loss of air pressure and water flow alarm signal operates properly.

8. For final commissioning, reset the system to operation status and advise all necessary parties that the system has been placed into service.

System placed and left in service:

Commissioning Completed By: _____
(Signature)

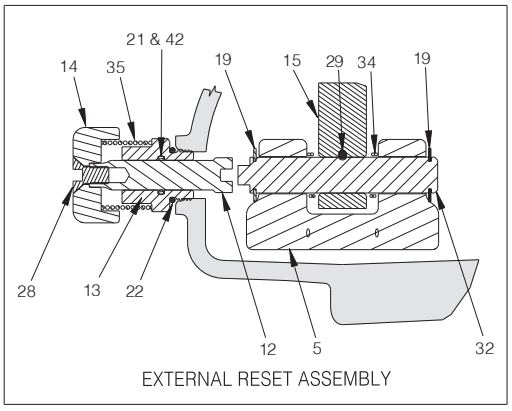
Date: _____

Representing: _____
(Installing Contractor)

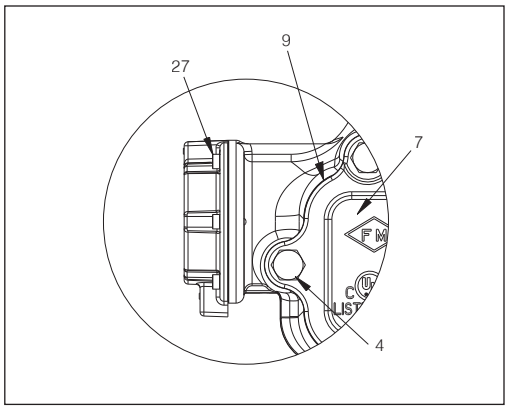
Notes:

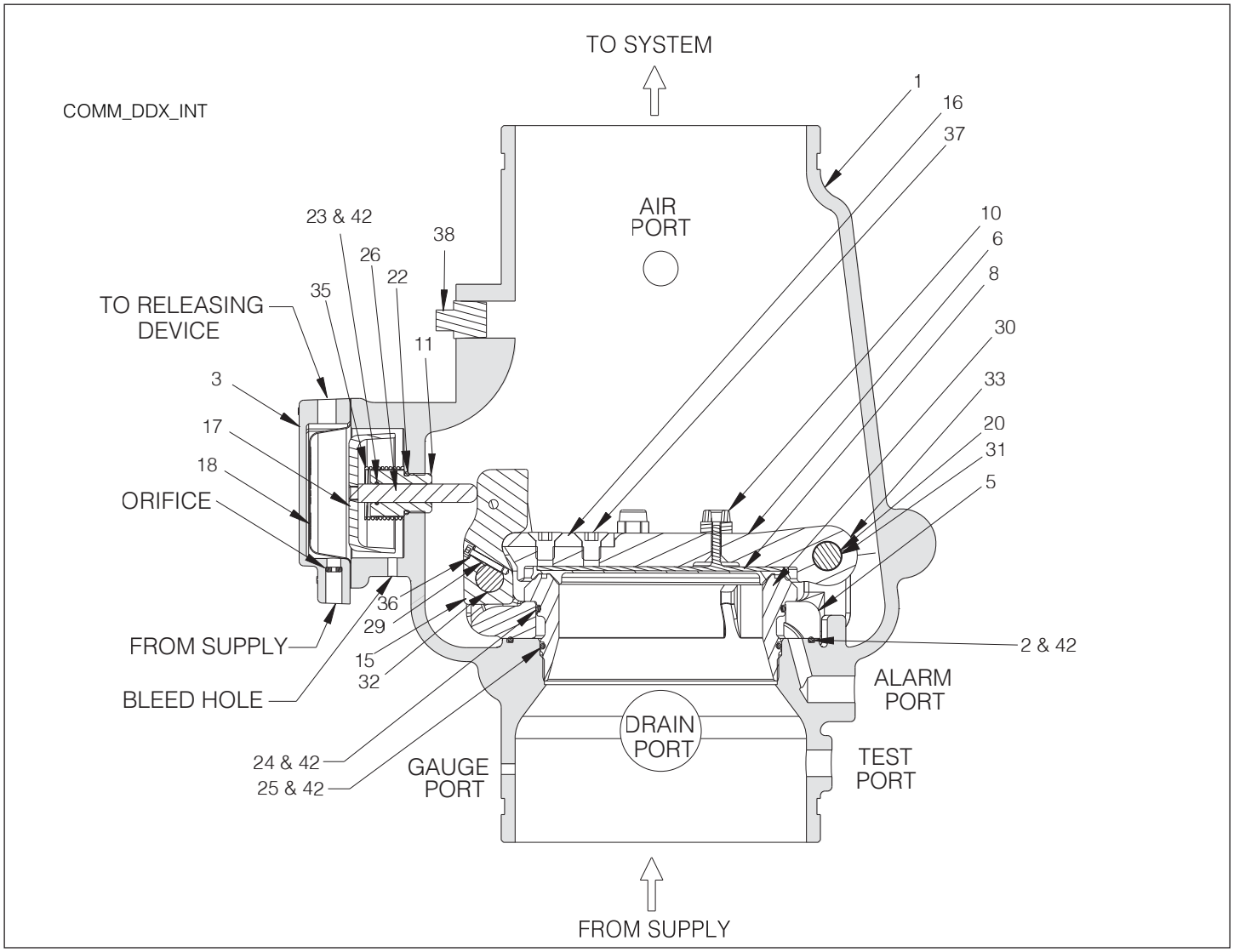
Annex A2

Model DDX Deluge Valve Components



EXTERNAL RESET ASSEMBLY





Annex A3

Model DDX Deluge Parts List

Item No.	Part No.								Part Description	QTY.	Material
	2" (50mm)	2-1/2" (65mm)	76mm	3" (80mm)	4" (100mm)	165mm	6" (150mm)	8" (200mm)			
1	91006011	91006012	91006023	91006013	91006005	91006027	91006007	91006028	Valve Body Groove/ Groove	1	Ductile Iron 65-45-12
	N/A	N/A	N/A	N/A	91006045	N/A	91006067	N/A	Valve Body Flange/ Groove		
	N/A	N/A	N/A	N/A	91006035	N/A	91006037	91006039	Valve Body Flange/ Flange		
2	N/A	N/A	N/A	N/A	N/A	N/A		95406414	O-ring (Mounting Ring)	1	Buna-N
3	71040416								Pushrod Cover Assembly	1	Ductile Iron 65-45-12 & Brass C360000
4	91106123				N/A	N/A		N/A	Hex Bolt ½"-13 x 1¼"	6	Zinc Plated Steel
	N/A				95606107	N/A		N/A	Hex Bolt ½"-13 x 1½"	6	
	N/A				N/A	91106006		N/A	Hex Bolt 5/8"-11 x 1¾"	6	
	N/A				N/A	N/A		95606110	Hex Bolt 5/8"-11 x 2"	8	
5	91306013				91306014	91306016		91306018	Mounting Ring	1	Stainless Steel CF8 or CF8M
6	91916003				91916014	91916016		91916008	Clapper	1	Stainless Steel CF8 or CF8M
7	92116063				92116064	92116065	92116066	92116068	Access Cover	1	Ductile Iron 65-45-12
8	93416003				93416014	93416016		93416008	Seal Assembly	1	Stainless Steel 304 & EPDM
9	93706003				93706004	93706006		93706008	Access Cover Gasket	1	Buna-N or Neoprene
10	93722000				93722000	N/A		N/A	Bumpstop Assembly	1	Stainless Steel UNS S31600 & EPDM
	N/A				N/A	93722000		N/A		2	
	N/A				N/A	N/A		93722000		3	
11	93916006								Pushrod Guide	1	Acetal
12	93916066								Reset Shaft	1	Brass UNS C36000
13	94106066								Reset Housing	1	Brass UNS C36000
14	94356006								Reset Knob	1	Aluminum 6061
15	94506003				94506004	94506016		94506008	Lever	1	Stainless Steel UNS S17400
16	95006412				94006412	95006410		95006410	Striker	1	Aluminum Bronze C95400

Annex A3 (cont.)

Model DDX Deluge Parts List

Item No.	Part No.								Part Description	QTY.	Material
	2" (50mm)	2-1/2" (65mm)	76mm	3" (80mm)	4" (100mm)	165mm	6" (150mm)	8" (200mm)			
17	95106006								Piston	1	Stainless Steel CF8M
18	95276006								Diaphragm	1	EPDM & Polyester
19	95306267				N/A	N/A		N/A	Retaining Ring, 3/8" Shaft, Lever Pin	2	Stainless Steel 15-7 or 17-7
	N/A				95306267	N/A		N/A	Retaining Ring, ½" Shaft, Lever Pin		
	N/A				N/A	95306269		N/A	Retaining Ring, 5/8" Shaft, Lever Pin		
	N/A				N/A	N/A		95316408	Retaining Ring, ¾" Shaft, Lever Pin		
20	95306268				N/A	N/A		N/A	Retaining Ring, 3/8" Shaft, Hinge Pin	2	Stainless Steel 15-7 or 17-7
	N/A				95306267	95306267		N/A	Retaining Ring, ½" Shaft, Hinge Pin		
	N/A				N/A	N/A		95316408	Retaining Ring, 3/4" Shaft, Hinge Pin		
21	95406007								O-Ring, Reset Housing ID	1	Buna-N
22	95406024								O-Ring, Reset Housing & Pushrod Guide OD	2	Buna-N
23	95406407								O-Ring, Pushrod Guide ID	1	Buna-N
24	95406410				95406409	95436126		95406413	O-Ring, Upper Seat	1	Buna-N
25	95406411				95406420	95446226		95406412	O-Ring, Lower Seat	1	Buna-N
26	95506006								Pushrod	1	Stainless Steel UNS S30300
27	95606114								Socket Head Screw, ¼"-20 x 5/8"	6	Steel
28	95606127								Flat Head Socket Cap Screw 3/8"-16 x ¾"	1	Steel
29	95606133				N/A	N/A		N/A	Socket Head Screw #6-32 x ½"	1	Stainless Steel 18-8
	N/A				95606130	95606130		95606130	Socket Head Screw #10-32 x 1"		Stainless Steel UNS S31600
30	96016003				96016014	96016016		96016008	Seat	1	Stainless Steel CF8M
31	96206003				N/A	N/A		N/A	Hinge Pin	1	Stainless Steel UNS S30400
	N/A				96216086	96216086		96206008			Stainless Steel UNS S21800

Annex A3 (cont.)

Model DDX Deluge Parts List

Item No.	Part No.								Part Description	QTY.	Material
	2" (50mm)	2-1/2" (65mm)	76mm	3" (80mm)	4" (100mm)	165mm	6" (150mm)	8" (200mm)			
32	96216003				N/A	N/A		N/A	Lever Pin	1	Stainless Steel UNS S17400
	N/A				96216044	96216047		96216008			Stainless Steel UNS S21800
33	96310003				96906904	96906904		96310008	Clapper Spacer	2	Teflon or Acetal
34	96406003				N/A	N/A		N/A	Lever Spring	1	Stainless Steel UNS S30400
	N/A				96406004	96406005		96406008			Stainless Steel UNS S31600
35	96406906								Piston/ Reset Spring	2	Stainless Steel UNS S31600
36	96906112				N/A	N/A		N/A	Spring Lock Washer, #6	1	Stainless Steel 18-8
	N/A				96906111	96906111		96906111	Spring Lock Washer, #10		Stainless Steel UNS S31600
37	95606140				N/A	N/A		N/A	Flat Head Socket Cap Screw ¼"-20 x ½"	2	Stainless Steel 18-8
	N/A				95606139	N/A		N/A			Stainless Steel UNS S31600
	N/A				N/A	N/A		95606135	Flat Head Socket Cap Screw ½"-13 x ¾"		Stainless Steel UNS S31600
38	98604402								Plug, ½" NPT	1	Steel
39	94616921								Knob Caution Label (Not Shown)	1	Polystyrene
40	91556922								Ball Chain, 1/8" (Not Shown) (Length is in Inches)	6	Nickel Plated Brass
41	91556923								Clamping Link, Ball Chain (Not Shown)	1	Nickel Plated Brass
42	6999993406								O-Ring Grease, Duponttm Krytox® GPL-205	A/R	Krytox®